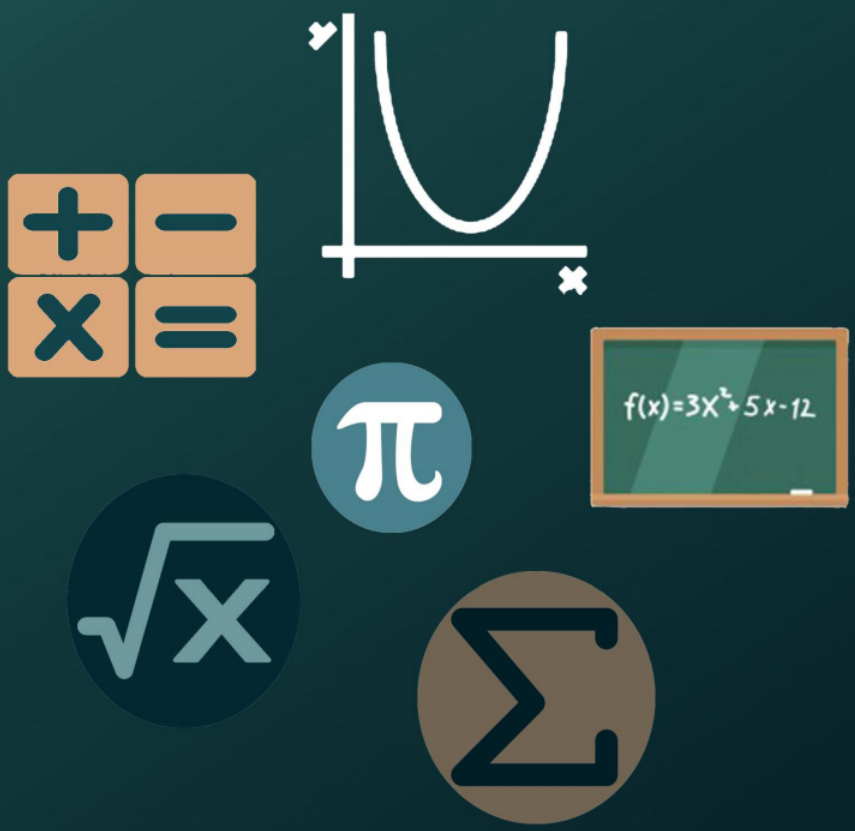
MATHEMATICS

Mathematical Reasoning



SHEET

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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr. No.	Topics	Ex-1	Ex-2
1	Logical Operator		
2	Logical Operator : Conditional and Biconditional	1-6	1-6
3	Algebra of statements	7-10	7-18

Exercise - 1 Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

1. The inverse of the statement $(p \wedge \sim q) \rightarrow r$ is-

(A) $\sim(p \vee \sim q) \rightarrow \sim r$

(B) $(\sim p \wedge q) \rightarrow \sim r$

(C) $(\sim p \vee q) \rightarrow \sim r$

(D) None of these

2. $(\sim p \vee \sim q)$ is logically equivalent to-

(A) $p \wedge q$

(B) $\sim p \rightarrow q$

(C) $p \rightarrow \sim q$

(D) $\sim p \rightarrow \sim q$

3. The equivalent statement of $(p \leftrightarrow q)$ is-

(A) $(p \wedge q) \vee (p \vee q)$

(B) $(p \rightarrow q) \vee (q \rightarrow p)$

(C) $(\sim p \vee q) \vee (p \vee \sim q)$

(D) $(\sim p \vee q) \wedge (p \vee \sim q)$

4. If the compound statement $p \rightarrow (\sim p \vee q)$ is false then the truth value of p and q are respectively-

(A) T, T (B) T, F

(C) F, T (D) F, F

5. The statement $(p \rightarrow \sim p) \wedge (\sim p \rightarrow p)$ is-

(A) a tautology

(B) a contradiction

(C) neither a tautology nor a contradiction

(D) None of these

6. Negation of the statement $(p \wedge r) \rightarrow (r \vee q)$ is-

(A) $\sim(p \wedge r) \rightarrow \sim(r \vee q)$

(B) $(\sim p \vee \sim r) \vee (r \vee q)$

(C) $(p \wedge r) \wedge (r \wedge q)$

(D) $(p \wedge r) \wedge (\sim r \wedge \sim q)$

7. The dual of the statement $\sim p \wedge [\sim q \wedge (p \vee q) \wedge \sim r]$ is-

(A) $\sim p \vee [\sim q \vee (p \vee q) \vee \sim r]$

(B) $p \vee [q \vee (\sim p \wedge \sim q) \vee r]$

(C) $\sim p \vee [\sim q \vee (p \wedge q) \vee \sim r]$

(D) $\sim p \vee [\sim q \wedge (p \wedge q) \wedge \sim r]$

8. Which of the following is correct-

(A) $(\sim p \vee \sim q) \equiv (p \wedge q)$

(B) $(p \rightarrow q) \equiv (\sim q \rightarrow \sim p)$

(C) $\sim(p \rightarrow \sim q) \equiv (p \wedge \sim q)$

(D) $\sim(p \leftrightarrow q) \equiv (p \rightarrow q) \vee (q \rightarrow p)$

9. The contrapositive of $p \rightarrow (\sim q \rightarrow \sim r)$ is-

(A) $(\sim q \wedge r) \rightarrow \sim p$

(B) $(q \rightarrow r) \rightarrow \sim p$

(C) $(q \vee \sim r) \rightarrow \sim p$

(D) None of these

10. The converse of $p \rightarrow (q \rightarrow r)$ is-

(A) $(q \wedge \sim r) \vee p$

(B) $(\sim q \vee r) \vee p$

(C) $(q \wedge \sim r) \wedge \sim p$

(D) $(q \wedge \sim r) \wedge p$

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

1. Write the negation of the statement if $5>7$ then, $6<4$.

(A) $5>7\wedge(6\geq 4)$ (B) $5>7\vee(6\leq 4)$

(C) $5\leq 7\vee(6\leq 4)$ (D) $5\leq 7\wedge(6\leq 4)$

2. The statement $\sim(p\leftrightarrow\sim q)$ is

(A) a tautology

(B) a fallacy

(C) equivalent to $p\leftrightarrow q$

(D) equivalent to $\sim p\leftrightarrow q$

3. Which one of the following is not a contradiction

(A) $\left[\sim p\wedge(p\vee\sim q)\right]\wedge q$

(B) $\sim(p\wedge q)\wedge p$

(C) $\left[(p\rightarrow q)\rightarrow p\right]\wedge\sim p$

(D) $(\sim p\rightarrow\sim q)\leftrightarrow(p\rightarrow q)$

4. Write down the negation of the statement it is raining and is cool.

(A) it is not raining and it is cool

(B) it is raining and it is cool

(C) neither it is raining nor it is cool.

(D) it is raining and it is not cool

5. $\left[p\wedge(\sim q)\right]\wedge\left[(\sim p)\vee q\right]$ is -

(A) a tautology

(B) a contradiction

(C) a tautology and a contradiction

(D) neither a tautology nor a contradiction

6. The statement $p\wedge(\sim q\vee r)\vee(\sim r\vee q)$ is -

(A) tautology

(B) contradiction

(C) contrapositive

(D) neither a tautology nor a contradicion

7. Let S be a non-empty subset of R. Consider the following statement :

P : There is a rational number $x\in S$ such that $x>0$. Which of the following statements is the negation of the statement P ?

(A) There is no rational number $x\in S$ such that $x\leq 0$

(B) Every rational number $x\in S$ satisfies $x\leq 0$

(C) $x\in S$ and $x\leq 0\Rightarrow$ is not rational

(D) There is a rational number $x\in S$ such that $x\leq 0$

8. $(p\Rightarrow q)\rightarrow\left[(r\vee p)\Rightarrow(r\vee q)\right]$ is -

(A) a contradiction

(B) a tautology

(C) a tautology and a contradiction

(D) neither a tautology nor a contradiction

9. Find the truth set of the open statement $p(x)\equiv x^2-x\left(3+\sqrt{2}\right)+3\sqrt{2}=0\ \ \forall x\in Q$.

(A) $\left\{\sqrt{2}\right\}$ (B) $\{3\}$ (C) $\left\{3,\sqrt{2}\right\}$ (D) $\{\ \}$

10. Suppose p: A natural number n is odd and q: natural number n is not divisible by 2, then the biconditional statement $p\Leftrightarrow q$ is -

(A) A natural number n is odd, if and only if it is divisible by 2

(B) A natural number n is odd, if and only if it is not divisible by 2.

(C) If a natural number n is odd, then it is not divisible by 2

(D) A natural number n is odd, then it is divisible by 2.

11. Write negation of the statement if ΔABC is isosceles then the base angles A and B are equal.

(A) ΔABC is isosceles and the base angle A and B are not equal

(B) ΔABC is isosceles and the base angle A and B are equal

(C) ΔABC is not isosceles and the base angle A and B are not equal

(D) all the above

12. The statement $p \to (q \to p)$ is equivalent to -

(A) $p \to (p \to q)$ (B) $p \to (q \vee p)$

(C) $p \to (q \wedge p)$ (D) $p \to (p \leftrightarrow q)$

13. Write the converse of the statement if $x+2=6$, then $x=4$

(A) If $x \neq 4$, then $x+2=6$

(B) If $x=4$ and $x+2=6$

(C) If $x+2=6$ then $x=4$

(D) If $x=4$, then $x+2=6$

14. If p, q, r have truth values T, F, T respectively, which of the following is true ?

(A) $(p \to q) \wedge r$ (B) $(p \to q) \wedge \square r$

(C) $(p \wedge q) \wedge (p \vee r)$ (D) $q \to (p \wedge r)$

15. Which of the following is always true?

(A) $(p \to q) \cong (\sim q \to \sim p)$

(B) $\sim (p \vee q) \cong (\sim p \vee \sim q)$

(C) $\sim (p \to q) \cong (p \vee \sim q)$

(D) $\sim (p \wedge q) \cong (\sim p \wedge \sim q)$

16. The negation of the statement "If I become a teacher, then I will open a school" is -

(A) I will become a teacher and I will not open a school

(B) Either I will not become a teacher or I will not open a school

(C) Neither I will become a teacher nor I will open a school

(D) I will not become a teacher or I will open a school.

17. The contrapositive and converse of the statements "if the two lines are parallel, then they do not intersect in the same plane" is

(A) If the two straight lines intersect in a plane, then the lines are not parallel. If the two lines do not intersect in the same plane, then the two lines are parallel

(B) If the two straight lines intersect in a plane, then the lines are parallel. If two lines do not intersect in the same plane, then the two lines are parallel.

(C) If two straight lines intersect in a plane, then the lines are not parallel. If the two lines do not intersect in the same plane, then the two lines are not parallel.

(D) If the two lines do not intersect in the same plane, then the two lines are not parallel. If two straight lines intersect in a plane, then the lines are not parallel.

18. $(p \Rightarrow q) \wedge (q \Rightarrow r) \Rightarrow (p \Rightarrow r)$ is -

(A) a tautology

(B) a contradiction

(C) a tautology and a contradiction

(D) neither a tautology nor a contradiction

Answer Key

Ex - 1									
1	2	3	4	5	6	7	8	9	10
C	C	D	B	B	D	C	B	A	A

Ex - 2									
1	2	3	4	5	6	7	8	9	10
A	C	B	C	B	D	B	B	B	B
11	12	13	14	15	16	17	18		
A	B	D	D	A	A	A	A		

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